

Dakshi Goel

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Education

Indian Institute of Technology(IIT), Jammu, India

B.Tech., Electrical Engineering (Minor Degree - Computer Science)

July, 2020 – May, 2024

GPA - 8.53/10.0

Relevant Coursework: DBMS, DSA, ML, Deep Learning, AI, Calculus, Linear Algebra

Delhi Public School, Vasant Kunj, India

Higher Secondary Standard

May, 2018 - May, 2020

94.8%

Skills

Programming Skills: Python, C++, Pandas, Numpy, Matplotlib, Sklearn, Tensorflow, Keras, PyTorch, FastAPI, REST API, BeautifulSoup, MySQL,

Technical Skills: RAGs, LLMs, Gen AI, NLP, Machine Learning, SQL, Data Visualization, Computer Vision, Probability, Jupyter, LangChain, Llama-Index, OpenAI, HuggingFace, Forecasting, Deep Learning.

Work Experience

Conversational AI Developer - Mozrest, Germany

Dec, 2023 - Current

- Build an AI review reply assistant using OpenAI endpoint, handled different types of reviews and ratings using prompt engineering and linked it with the restaurant data from SQL database.
- Created a random forest algo for the booking slot availability prediction with data more than 3k features and achieved an accuracy of 94% above a confidence score of 90%. Classified the restaurant on the basis of long. and lat., and categorised the ratings and price level, using the GMM and KNN model into 5 clusters.

AI and Innovation Intern - Infineon, Bangalore

Jan, 2024 - June 2024

- Developed a chatbot application using LLamaIndex for the task of long text-summarization using the method for recursive chunking and build a RAG pipeline for question-answering by parsing advanced pdf and url.
- Deployed a Nocode-Lowcode website on the internal cloud platform with integrated genai using REST API,for applications like text-translation, and personalization tasks for the technical documentations.

Project Intern - LTIMindtree, Bangalore, India

May, 2023 - July, 2023

- Optimized the performance and speed of the LLM (Large Language Model) based on the Stanford Alpaca model, using Llangchain, Hugging Face and Llama-index.
- Performed web-scraping and chunking using BeautifulSoup and used Llangchain for the fine tuning of the open-Llama model. Created an interactive chatbot user interface by employing Streamlit.

Research Intern - University of New South Wales, Sydney, Australia (Remote)

June, 2022 - Dec, 2022

- Addressed class imbalance in ML/DL datasets by combining two different techniques of SMOTE and GANs, achieving about 2-3% improvement in F1 scores for classification tasks on oversampled test datasets.
- Trained and optimized CNN and MLP models for predicting alteration zones, a geological feature, using hyperspectral remote sensing data, achieving the highest F1 score of 0.91, beating the benchmark ML models.

ML Intern - UNIV.AI

Feb, 2021 - April, 2021

- Composed 10+ detailed and interactive Jupiter Notebooks for the coursework targeting Data Science enthusiasts on topics -Statistics, Data manipulation, and Exploratory Data Analysis.
- Worked on multiple datasets, performed EDA, calculated the statistical data, and implemented python libraries like Numpy, Pandas, Matplotlib, Scikit-learn, Seaborn, and PyTorch.

Academic Projects

Underwater Image Enhancement: IIT Jammu

Jan, 2024 - Current

- Developed a new multi-scale U-Net architecture, outperforming existing methods by 1.18 dB in PSNR, with visually appealing and color-accurate results.
- Innovated novel techniques like multi-head attention fusion modules and different losses based in color models to improve the performance.

Flipkart Grid Challenge 5.0 (Software)

Nov 2023 - Nov, 2023

- Fine-tuned Llama model on synthetically generated data and developed a RAG pipeline using fashion product recommendations based on scrapped data from fashion vlogs and Instagram to capture current trends.
- Integrated the RAG pipeline with text-to-SQL llm model to fetch data from the SQL database for personalization based on customer buying history.

Age and Gender Prediction, Inter-IIT TechMeet, IIT KGP

Dec, 2021 - Dec, 2021

- Developed an image processing pipeline using super-resolution techniques like ESRGANs and Bilinear Interpolation to improve the quality of detected images.
- Implemented a novel technique of weighted average of Age Classification and Regression models based on confidence score to estimate a person's age.

Publications

Remote sensing data processing using convolutional neural networks for mapping alteration zones accepted to be published in the IEEE Xplore Digital Library.

A system and a method fir underwater image enhancement filed at Indian patent office (No. 202411053490) and under consideration for publication at IEEE Transactions on Image Processing as a Regular Paper (S1).

Achievements

- Won **second prize** in Pitch deck competition SEED by MIET, in the region of Jammu.
- **First Runner's Up** in Flipkart Grid Robotics Challenge 3.0. Over 9000 teams participated in pan India.
- Ranked 4th in model accuracy and code quality in Inter-IIT Tech Meet out of the 23 top IITs.
- Selected in the top 5 teams among 75 teams in pan India by presenting an Innovative solution in Hackathon organized jointly by SERB-INAE, Jadavpur University.

Position of Responsibility

- Student Head Representative - Career Developement Services, IIT Jammu
- Dance Club Coordinator, IIT Jammu